

SafeRoute AI — Dynamic Fire Evacuation Router

Decentralized Edge Mesh with Real-Time Physical Hazard Mapping & 3D Digital Twin

Problem Statement ID: SIH-2026-SR01 (Dynamic Fire Evacuation Router)

Problem Statement Title: Dynamic Fire Evacuation Router with Real-Time Hazard Mapping

Theme: Smart Life-Safety Infrastructure & Edge AI Systems

PS Category: Hardware & Embedded Software Integration

Team Name / Leads: SafeRoute AI Core Engineering Team

Key Innovations: Sub-300ms Edge Dijkstra • ESP-NOW Mesh • 3-Tier Fail-Safe • 3D EOC Twin

PROPOSED SOLUTION

PROPOSED SOLUTION / PROTOTYPE

Decentralized Edge Graph

Each node acts as an OSPF-style physical router running Dijkstra on-device.

Multi-Vector Sensor Fusion

Fuses Temperature (°C), Smoke (PPM), Flame (IR), and Occupancy into dynamic edge weights.

Dynamic Visual Guidance

Drives WS2812B chasing LED strips to direct occupants away from hazards.

HOW IT ADDRESSES THE PROBLEM

Eliminates Static Trap

Recalculates paths instantly when fire blocks an exit, reversing chasing LED direction.

Sub-300ms Reaction Time

Updates visual indicators before occupant panic commits people to danger zones.

Zero Single-Point-of-Failure

Runs 100% autonomously without reliance on central servers or power mains.

INNOVATION & UNIQUENESS

Continuous Exponential Math

Uses exponential cost functions $\exp(\alpha T + \beta S)$, avoiding crude binary thresholds.

Lock-Free FreeRTOS Core

Core 0/1 double-buffering avoids mutex priority inversion on hot routing path.

3-Tier Fail-Safe Hierarchy

Seamlessly transitions from local sensors to neighbor consensus on hardware fault.

TECHNICAL APPROACH

TECHNOLOGIES & ARCHITECTURE

Embedded Hardware

ESP32 Dual-Core 240MHz, DHT22 (Temp), MQ-2 (Smoke), IR Flame, WS2812B LEDs, Buzzer.

Firmware Stack

C++17 under PlatformIO / FreeRTOS. Core 1 handles ESP-NOW, Core 0 runs Dijkstra.

Networking Mesh

ESP-NOW 2.4GHz connectionless flooding with 24-byte packet, CRC16, and monotonic seq-nums.

Simulation & Twin

Python digital twin injector tool, FastAPI backend bridge, Three.js 3D WebGL EOC UI.

MATHEMATICAL FORMULA & LED MATRIX

$$\text{edge_cost} = (\text{base_dist} \cdot \exp(2.2 \cdot T_{\text{norm}} + 1.6 \cdot S_{\text{norm}}) + 0.5 \cdot 0_{\text{norm}} \cdot \text{base_dist}) \cdot (\text{Flame} ? 10^6 : 1)$$

GREEN

Safe shortest path; chasing lights point to exit.

YELLOW

Rerouted alternate route (Smoke-dominant: $S_{\text{norm}} > T_{\text{norm}}$).

PULSING RED

Immediate danger (Active flame OR $T_{\text{norm}} \geq S_{\text{norm}}$).

WHITE STROBE

Shelter-in-place state (All exit costs $\geq 100,000$).

FEASIBILITY AND VIABILITY

FEASIBILITY ANALYSIS

Hardware Cost (\$16-\$24)

Commodity ESP32 nodes and standard sensors keep per-node BOM extremely low.

Retrofit Friendly

Installs alongside existing code-mandated static signage without rewiring.

Zero Cloud Cost

Operates 100% autonomously without requiring expensive cloud subscriptions.

POTENTIAL CHALLENGES & RISKS

Sensor Failure / Stuck ADC

Physical sensors can break, short-circuit, or freeze ADC values during fire conditions.

Packet Corruption / Drop

RF interference or fire damage can drop mesh packets or corrupt data payloads.

All Exit Traversal Blocked

Severe flashover can block all available building exit paths simultaneously.

MITIGATION STRATEGIES

3-Tier Fail-Safe Hierarchy

Local variance check detects stuck sensors; automatically defers to Tier 2 Neighbor Consensus.

CRC16 & Monotonic Seq-Num

Validates payload integrity and rejects replayed or corrupted packets live.

Shelter-In-Place State

Detects cost $\geq 100,000$, triggering White Strobe LEDs to instruct room sealing.

ARTIFACTS

FIRMWARE & ALGORITHM CODE

- **Modular Repository:** firmware/src/ (main, routing, fusion, comms, leds, failsafe).
- **On-Device Dijkstra:** Non-blocking link-state graph solver running on ESP32 Core 0.
- **Double-Buffer Swap:** Lock-free atomic pointer swap synchronization.

FIRE INJECTOR SIMULATOR

- **Multi-Profile Injector:** simulator/injector.py streams smolder & flashover profiles.
- **Judge Attack Trigger:** Allows judges to trigger flashover on arbitrary zones live.
- **Corrupt Packet Generator:** Injects malformed CRC payloads to prove live rejection.

3D EOC DIGITAL TWIN

- **Three.js WebGL Interface:** dashboard/ renders 2D/3D floor grid & exit paths live.
- **Async IDW Heatmap:** Inverse Distance Weighting interpolates heat/smoke surfaces.
- **Node Health Panel:** Surfaces sensor fault alerts and consensus state live.

PHYSICAL TESTBENCH

- **Hardware Prototypes:** ESP32 nodes wired with WS2812B strips, MQ-2, DHT22.
- **Wokwi Simulation:** Full hardware binary simulation suite under tests/.
- **Timing Benchmark Logs:** Measured p95 reaction latency of 118ms across 4 hops.

RESEARCH AND REFERENCES

DATASETS & CURVE REGRESSION

NIST Fire Dynamics Data

Extracted temperature rise timelines and smoke flashover curves from National Institute of Standards and Technology public research.

Kaggle Smoke Datasets

Used public smoke detection time-series data to fit logistic growth curves for smoldering fire profiles.

Hyperparameter Tuning

Offline regression scripts under simulator/fire_profiles/ fit $\alpha=2.2$ (thermal) and $\beta=1.6$ (smoke) parameters.

STANDARDS & TECHNICAL REFERENCES

NFPA 101 Life Safety Code

Framed system to augment — not replace — code-mandated static illuminated exit signage in commercial facilities.

OSPF RFC 2328 Protocol

Applied classical link-state network graph routing concepts to physical hazard topology modeling.

FreeRTOS Dual-Core Architecture

Implemented lock-free double-buffered pointer swapping based on real-time embedded systems literature.